Department of Physics

PCS 125 – Waves and Fields

Please insert in the circle the initial of your last name inside

Please Print Clearly:

Student:_____ _____

Last Name ***First Name***

Student Number: _____ **Student Section:** _____

Final exam – April 25, 2012

Check the blank box in from of the name of your professor:

Dr. Carvalho		Dr. Chithrani		Dr. Tavakkoli		Dr. Yuan	

Read the following attentively

- Duration of the exam: 3 hours (180 minutes).
- The exam has 14 pages (including the front page and the formula sheet page).
- The exam consists of 26 multiple choice questions, each worth one mark. The maximum exam mark is 26.
- Note that for each multiple choice question:
  - You have to provide, in this booklet, a justification for your answer.
  - The work shown in your exam booklet may be checked. If no work or correct argument is shown to validate your answer, the resulting mark may be zero, even if the answer in the scantron is correct.
  - Transfer your choices to the scantron only when you're sure of them. Some answers may be rounded off.
- All Ryerson exam rules apply.
- Sharing of pens, erasers and calculators is not permitted.
- Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.
- A formula sheet is provided with the test booklet. No other sheets are allowed.
- The only calculators allowed are either the Sharp EL-546 or Casio FX-991. If you are found to have another type of calculator (even if not programmable) it will be confiscated for the duration of the test

_____

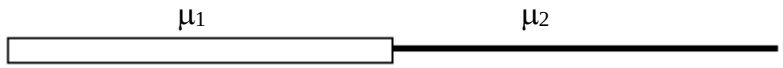
Student signature (By signing here you indicate that you read all the instructions. No signature, no mark!)

Question 1.
What do we need to change in order to double the total energy of a mass oscillating at the end of a spring?

(a) increase the angular frequency by $\sqrt{2}$
(b) increase the amplitude by $\sqrt{2}$.
(c) increase the amplitude by 2.
(d) increase the angular frequency by 2.
(e) increase the amplitude by 4 and decrease the angular frequency by $\frac{1}{\sqrt{2}}$.

Question 2.
A string attached at one end to a vibrator of fixed frequency is subjected to varying tensions. When the tension is 20 N, a resonance with 3 antinodes (loops) results. What tension would cause a resonance with 2 antinodes in the same length of this string?
(a) 30 N
(b) 45 N
(c) 80 N
(d) 8.9 N
(e) 13 N

Question 3.
The figure below represents a string of a given material, under a fixed tension T, which is made of two sections of different thicknesses. It is know that $\mu_1=16\mu_2$. What is the ratio v_1/v_2 where v_1 is the speed of a pulse traveling in the thicker section, and v_2 the speed of that same pulse traveling in the other section of the string?

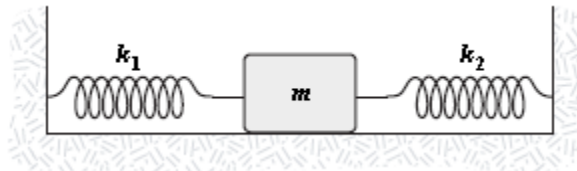


(a) $\frac{1}{16}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) 8 (e) 1

Question 4.
Suppose a grandfather clock is calibrated correctly at sea level and is then taken to the top of a very tall mountain. Does the grandfather clock run

(a) slow
(b) fast
(c) correctly
(d) slow or fast, depending on the height of the mountain.
(e) slow or fast, depending on the mass of the mountain.

Question 5.
Observe the figure given in this question. The block of mass *m* slides on a frictionless surface. If m = 2 kg, k_1 = 800 N/m and k_2 = 500 N/m, what is the natural frequency of the oscillations?



(a) 25 Hz

(b) 2 Hz

(c) 4 Hz

(d) 8 Hz

(e) 10 Hz

Question 6.

For the transverse wave described by $y = 0.15\sin\left[\frac{\pi}{16}(2x - 64t)\right]$ where *x* and *y* are in meters and *t* in seconds. Determine the maximum transverse speed of the particles of the medium.

(a) 0.192 m/s

(b) 0.6π m/s

(c) 9.6 m/s

(d) 4π m/s

(e) 32 m/s

Question 7.

Two harmonic waves traveling in opposite directions interfere to produce a standing wave described by $y = 2 \sin(\pi x) \cos(3\pi t)$ where x is in meters and t is in seconds. What is the distance between two consecutive antinodes?

(a) 8 m

(b) 2 m

(c) 4 m

(d) 1 m

(e) Other (give value)

Question 8.

A point source emits sound waves with a power output of 100 watts. What is the sound level (in dB) at a distance of 10 m?

(a) 109

(b) 119

(c) 129

(d) 103

(e) other (give value)

Question 9.

A tube in air (20°C) closed at one of its ends supports standing waves at frequencies 850 Hz and 1190 Hz and none in between. The second lowest standing wave possible in this tube has frequency

(a) 150 Hz

(b) 200 Hz

(c) 510 Hz

(d) 750 Hz

(e) other (give value)

Question 10.

An automobile's horn produces a frequency of 780 Hz. How fast is the car traveling if a stationary microphone measures the horn's frequency to be 863.8 Hz. The temperature of air is 28.8^{o}C on that day.

(a) 120 km/h

(b) 200 km/h

(c) 110 km/h

(d) 85.0 km/h

(e) other (give value)

Question 11.

A rock is dropped from a distance R_E above the surface of the earth, and is observed to have kinetic energy K_1 when it hits the ground. An identical rock is dropped from twice the height ($2R_E$) above the earth's surface and has kinetic energy K_2 when it hits the ground. R_E is the radius of the earth. What is K_2/K_1?

(a) 2

(b) 4/3

(c) 3/2

(d) 2/3

(e) 4

Question 12.

A +20 nC point charge is placed on the x axis at x = 2.0 m, and a -25 nC point charge is placed on the y axis at y = -3.0 m. What is the angle that the direction of the electric field at the origin makes with the x-axis?

(a) 209°

(b) 61°

(c) 29°

(d) 241°

(e) 119°

Question 13.

If Q = +14 nC, a = 3 m, and b = 4 m, what is the magnitude of the electric field at point P?

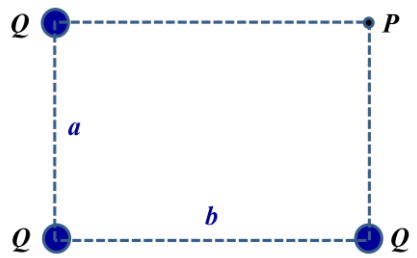
(a) 24 N/C

(b) 31 N/C

(c) 13 N/C

(d) 19 N/C

(e) 21 N/C



Question 14.

Four identical 2.0 μC charges are located on the vertices of a square with 2.0 m long sides. Determine the electric potential (relative to infinity) at the center of the square.

(a) 38 kV

(b) 23 kV

(c) 76 kV

(d) 64 kV

(e) 51 kV

Question 15.

A particle (of charge q = +4.0 mC and mass m = 50 g) has a velocity of 25 m/s in the positive x direction when it enters a region where a uniform electric field of magnitude E=60 N/C in the positive y direction exists. What is the speed of the particle 5 s after it enters this region? Neglect the deflection due to gravity.

(a) 49 m/s

(b) 35 m/s

(c) 32 m/s

(d) 44 m/s

(e) 24 m/s

Question 16.

A current of 4.0 A flows in a 4.0-Ω resistor during 5.0 min. How many electrons pass through the resistor during this time interval?

(a) 7.5×10^{21}

(b) 5.6×10^{21}

(c) 6.6×10^{21}

(d) 8.4×10^{21}

(e) Other (give value)

Question 17.

Which statement is correct?

(a) The electric force acts perpendicular to the direction of the electric field.

(b) The magnetic force acts along the magnetic field.

(c) The magnetic force acts on a charged particle regardless of whether the particle is moving or not.

(d) The electric force acts on a charged particle only when the particle is in motion.

(e) None of the above.

Question 18.

Point charges q and Q are positioned as shown. If q = +1.0 nC, Q = –2.0 nC, a = 3.0 m, and b = 4.0 m, what is the electric potential difference, V_A - V_B?

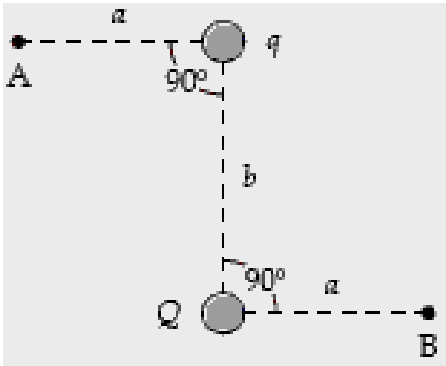
(a) 8.4 V

(b) 6.2 V

(c) 3.6 V

(d) 0 V

(e) 4.8 V



Question 19.

A singly charged ion of mass m_1 is accelerated from rest by a potential difference ΔV. It is then deflected by a uniform magnetic field (perpendicular to the ion's velocity) into a semicircle of radius R_1. Now a triply charged ion of mass m_2 is accelerated through the same potential difference and deflected by the same magnetic field into a semicircle of radius $R_2 = 2R_1$. What is the ratio m_2/ m_1 of the masses of the ions?

(a) 8

(b) 12

(c) 108

(d) 27

(e) 1

Question 20.

An electron has a velocity of 6.0×10^6 m/s in the positive x direction at a point where the magnetic field has the components $B_x = 3.0$ T and $B_y = 1.5$ T. What is the magnitude of the acceleration of the electron at this point?

(a) $2.1\ 10^{18}$ m/s^2

(b) $1.6\ 10^{18}$ m/s^2

(c) $2.6\ 10^{18}$ m/s^2

(d) $3.2\ 10^{18}$ m/s^2

(e) $3.5\ 10^{18}$ m/s^2

Question 21.

A 2.0 m long wire carries a current of 15 A directed along the positive *x* axis in a region where the magnetic field is uniform and given by **B** = (30 **i** – 40 **j**) mT. What is the resulting magnetic force on the wire?

(a) (+1.2 **k**) N

(b) (–1.2 **k**) N

(c) (–1.5 **k**) N

(d) (+1.5 **k**) N

(e) (+0.90 **k**) N

Question 22.

A Hall-effect probe operates with a 120 mA current. When the probe is placed in a magnetic field of magnitude 40 mT, it produces a Hall voltage of 0.800 μV. When it is used to measure an unknown magnetic field, the Hall voltage is 0.340 μV. What is the magnitude of the unknown magnetic field?

(a) 0.038 T

(b) 0.017 T

(c) 0.094 T

(d) 0.042 T

(e) other (give value)

Question 23.

A negatively-charged particle in an electric field

(a) gains potential difference and kinetic energy when it moves in the direction of the field.

(b) loses electric potential energy when it moves in the direction of the field.

(c) gains kinetic energy when it moves in the direction of the field.

(d) gains electric potential energy when it moves in the direction of the field.

(e) None of the above.

Question 24.

A particle of mass M and charge $Q>0$ moves in a region of space where the magnetic field has a constant magnitude B and a downward direction. What is the magnetic force on the particle at an instant when it is moving horizontally toward the north with a speed v?

(a) QvB toward the east

(b) zero

(c) QvB toward the west

(d) QvB upward

(e) QvB toward the south

Question 25.

Two long, parallel wires are attracted to each other by a force per unit length of 320 μN/m when they are separated by a vertical distance of 0.500 m. The current in the upper wire is 20.0 A to the right. Determine the current in the lower wire.

(a) 20.0 A.

(b) 40.0 A

(c) 33.0 A

(d) 15.0 A

(e) Other (give value)

Question 26.

Consider two concentric circular wires lying on the plane of this page. One of the wires has radius 2.00 cm and carries a current of 4 A in the clockwise direction; the other has radius 5.0 cm and carries a current of 10 A in the counterclockwise direction. What is the magnetic field at the center of circular wires?

(a) 0. 25 mT, into the page.

(b) 0. 25 mT, out of the page.

(c) 2.5 μT, out of the page.

(d) 2.5 μT, into the page.

(e) 0